

Shear Probe Calibration Report

PROBE DETAILS

Probe SN: M1458
Probe PN: 012-000-20
Description: Shear Probe, 3/8", SMC

CALIBRATION RESULTS

Sensitivity (sens or S): $0.0777 \frac{\text{V}}{\text{m}^2\text{s}^{-2}}$
Calibration Date: 2026/06/19
Recommended re-calibration: 2027/06/19
Previous Calibration Date: 2021/04/27
Previous Sensitivity: 0.0679

CALIBRATION METHOD

The shear probe is calibrated using a vertical jet of water at speed U . The angle of attack, α , is then varied from approximately -10° to $+10^\circ$ in steps of 2° and the output signal, V_{rms} , generated from the rotating probe is measured.

Least Squares Regression:

$$\frac{V_{rms}}{U^2} = S(0) \sin 2\alpha + c \sin^3 2\alpha$$

$$S(0) = 0.0777$$
$$c = 0.031886$$

Units:

$$[S(0), c] \equiv \frac{\text{V}}{\text{m}^2\text{s}^{-2}} \quad [\alpha] \equiv ^\circ$$

Your probes calibration data and results are shown on the next page.

TESTING DETAILS

Calibration Operator: AC
Calibration temperature: 14.9°C
Amplifier type: Charge Transfer
Measured Capacitance: 0.986 nF
Measured Resistance: $> 200.0 \text{ G}\Omega$
Shear DC Offset: -1.97 mV

PRESSURE TESTING DETAILS

Operator: SY
Date: 2016/04/19
Depth Rating: 6000 m



WARNING: THIS PROBE IS EXTREMELY FRAGILE!

Avoid contact between the probe tip and any solid object.

FREQUENTLY ASKED QUESTIONS

Q1) How often should shear probes be calibrated?

Frequent re-calibration is strongly encouraged to ensure an accurate sensitivity coefficient. Rockland recommends re-calibrating shear probes at least once per year, shortly before fieldwork, and ideally before and after fieldwork.

Q2) How to request a cold-water calibration and when is it done?

Cold water calibrations are performed at a calibration temperature of 5°C . Rockland recommends using a cold water calibration for deployments in environments where the water temperature averages 5°C or below. Please contact support@rocklandscientific.com to request a cold-water calibration.

Q3) What are the basic operating requirements of the shear probe?

To obtain measurements with an adequate signal-to-noise ratio a flow past the probe greater than 0.2 m s^{-1} is required. Please consult the instrument user manual for relative flow speed recommendations specific to the instrument and application. The angle of attack between the probe and this flow should not exceed 10° , and the probe should be mounted on a mechanically quiet platform.



If you have questions or you suspect a shear probe may be broken, please contact

support@rocklandscientific.com